<u>Proof</u> 2) We have $|Ric| < \delta$

$$\mathrm{Vol}\, B(p,1) \geq (1-\delta)\,\omega_n$$

ω_n = volume of $B(0,1) \subset \mathbb{R}^n$

Let $q \in B(p,\delta)$, $r \leq 1$

Bishop-Gromov $\Longrightarrow$

$$\frac{\mathrm{Vol}\, B(p,1)}{v(n,\delta,1+\delta)} \leq \frac{\mathrm{Vol}\, B(q,1+\delta)}{v(n,\delta,1+\delta)} \leq \frac{\mathrm{Vol}\, B(q,r\delta)}{v(n,\delta,r\delta)}$$

$$\Longrightarrow \mathrm{Vol}\, B(q,r\delta) \geq (1-\delta)\omega_n \cdot \underbrace{\frac{v(n,\delta,r\delta)}{v(n,\delta,1+\delta)}}_{\sim (r\delta)^n}$$

Scaling up by δ^{-1}: we have

- $|Ric| \leq \delta$
- $\mathrm{Vol}\, B(q,r) \geq (1-\Psi(\delta))\,\omega_n r^n$

  where $\Psi(\delta) \to 0$ as $\delta \to 0$

  $\forall\, q \in B(p,1)$, $r \leq 1$.

<u>Claim</u>: $r(p) \geq r_0$ for r_0 dep. on dimension if δ is sufficiently small

where $r(p) = \sup\{ r \leq 1 \mid \|(M,g,p)\|_{C^{1,\alpha},r} \leq 2 \}$

Argue by contradiction. If claim fails then we have sequence (M_i, g_i)

satisfying $|\mathrm{Ric}\, M_i| \le \delta_i \longrightarrow 0$

$$\mathrm{Vol}\, B(q,r) \geq (1-\Psi(\delta_i))\,\omega_n r^n$$

$$\forall q \in B(p_i,1),\ r \le 1$$

but $r(p_i) \to 0$ $\quad (p_i \in M_i)$

Let $q_i \in B(p_i,1)$ maximize

$$r(q_i)^{-1} d_{q_i}, \quad d_{q_i} = 1 - d(p_i, q_i)$$

we have $r(q_i)^{-1} d_{q_i} \longrightarrow \infty$

so $r(q_i)^{-1} \to \infty$

Consider rescaled pointed sequence

$$(\tilde{M}_i, \tilde{g}_i, \tilde{q}_i) \qquad \tilde{M}_i = M_i$$
$$\tilde{g}_i = r(q_i)^{-2} g_i$$
$$\tilde{q}_i = q_i$$

For any $q \in B(q_i, \frac{1}{2} d_{q_i})$

we have $r(q)^{-1} d_q \le r(q_i)^{-1} d_{q_i}$

$$\Rightarrow \quad r(q)^{-1} \le 2 r(q_i)^{-1}$$

i.e. for any $\tilde{q} \in B(\tilde{q}_i, \frac{1}{2} \underbrace{d_{q_i} r(q_i)^{-1}}_{\to \infty})$

we have $r(\tilde{q})^{-1} \le 2 r(\tilde{q}_i)^{-1} = 2$

$$r(\tilde{q}) \geq 1/2$$

$\Rightarrow$ we have uniform bounds

$$\| (\tilde{M}_i, \tilde{g}_i, \tilde{q}) \|_{C^{1,\alpha}_{,1/2}} \leq 2$$

for $\tilde{q}$ in larger and larger balls around $\tilde{q}_i$.

We can extract a pointed $C^{1,\alpha/2}$-limit

$$(\tilde{M}_i, \tilde{g}_i, \tilde{q}_i) \longrightarrow (M_\infty, g_\infty, q_\infty)$$

M_∞ is smooth Ricci flat and convergence is in $C^{1,\beta}$ $\forall \beta < 1$

For all $R > 0$ $B(q_\infty, R)$ is a $C^{1,\alpha}$-limit of $B(\tilde{q}_i, R)$

i.e. $B(q_i, r(q_i)R)$ scaled up by $r(q_i)^{-1}$

$r(q_i)R < 1$ for large i so

$$\mathrm{Vol}\, B(q_i, r(q_i)R) \geq (1 - \Psi(\delta_i)) \, \omega_n (r(q_i)R)^n$$

i.e. $\mathrm{Vol}\, B(\tilde{q}_i, R) \geq (1 - \Psi(\delta_i)) \omega_n R^n$

letting $i \to \infty$ we get

$$\mathrm{Vol}\, B(q_\infty, R) \geq \omega_n R^n$$

M_∞ is Ricci flat $\Rightarrow$ Bishop–Gromov implies

$$B(q_\infty, R) = B(0, R) \subset \mathbb{R}^n$$

isometrically

holds for all $R \Rightarrow M_\infty = \mathbb{R}^n$

we have maps

$$\varphi_i : B(0,R) \to \tilde{M}_i \quad \text{for large } i$$
$$\varphi_i(0) = \tilde{q}_i$$

and $\varphi_i^* \tilde{g}_i \longrightarrow g_{Euc}$ in $C^{1,\alpha}$

$$\Rightarrow \|(\tilde{M}_i, \tilde{g}_i, \tilde{q}_i)\|_{C^{1,\alpha},R} \to 0 \quad \text{as } i \to \infty$$

this contradicts $r(\tilde{q}_i) = 1$. $\square$

For Einstein metrics we get higher regularity:

<u>Prop</u> Suppose (M,g) is Einstein: $Ric(g) = \lambda g$, $|\lambda| \le 1$

and $\|(M,g)\|_{C^{1,\alpha},1} \le 1$.

Then $\|(M,g)\|_{C^{k,\alpha},r} \le C_k$ for

constants r, C_k depending on dimension n.

<u>Proof</u> Use the Einstein eq^n in harmonic coordinates. $\square$

<u>Corollary</u> $\exists\, \delta > 0$ (dep. on n) with following property: Let $\mathcal{C}$ be the class of Einstein mflds (M, g), $\mathrm{Ric}(g) = \lambda g$, $|\lambda| \leq 1$ satisfying $\mathrm{Vol}\, B(p, r) \geq (1-\delta)\, \omega_n r^n$ $(*)$ $\forall r < \delta$

Then $\mathcal{C}$ is compact in pointed $C^{k,\alpha}$ topology.

This fails if instead of $(1-\delta)$ in $(*)$ we put $v > 0$